

Question Paper

Exam Date & Time: 27-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Parallel Computer Architecture [NPT_4132]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain how architectural scalability is influenced by interconnection network topology in parallel systems. Compare mesh and fat-tree architectures in terms of latency and bandwidth utilization (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Evaluate the effectiveness of shared-memory versus message-passing models for parallel programming. Under what workload conditions does one outperform the other? (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Evaluate how virtual memory design influences parallel program scalability. Apply the different strategies to reduce TLB and page-fault overheads in high-core-count systems (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Explain how snooping and directory-based cache coherence protocols handle write operations differently. Illustrate the trade-off between scalability and coherence traffic in both approaches. (5)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- 3) Evaluate the effectiveness of the MESI and Dragon protocols for a multicore processor with high write-sharing workloads. Which protocol offers better performance and why? (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Propose an improved coherence strategy for a system using multi-level caches with an atomic bus. Justify your design choices with respect to latency, bandwidth, and consistency guarantees (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyze how integrating a split-transaction bus affects cache miss handling and bus arbitration in multi-level cache hierarchies. Discuss its impact on fairness and ordering. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Assess the correctness conditions (write serialization, data consistency, and absence of live lock) for a cache-based directory protocol. How do these properties ensure reliable shared-memory execution? (5)
- Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 5) Evaluate barrier synchronization techniques in shared-memory multiprocessors. How do hardware-based and software-based barriers differ in terms of latency and scalability? (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) Evaluate deterministic and adaptive routing strategies in interconnection networks. Propose a hybrid routing mechanism that minimizes deadlock while optimizing throughput. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4132:Parallel Computer Architecture

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	15
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4132:Parallel Computer Architecture

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(20%)
4	Analysis	4	20(40%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)